

A New Old Keynesian Model

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Abstract

The theoretical behavior of aggregate markups in New Keynesian models is partly at odds with their observed cyclicalities in the data, so that their explanatory power relies extensively on exogenous markup shocks. To reconcile theoretical predictions and empirical evidence, I develop a ‘New Old Keynesian’ model. This New Keynesian adaptation of the Barro-Grossman model of rationing-through-nominal-rigidities allows fix-price firms to ration excess demand when high marginal costs erode profitability. Rationing induces a forced reallocation of expenditure toward high-margin sectors, generating a first-order welfare loss from price dispersion.

I contend that rationing has the potential to improve the New Keynesian model from a business cycle accounting perspective. By approximating the policy functions with a deep neural network, I demonstrate that markups can be countercyclical with respect to welfare-relevant output while appearing procyclical with respect to measured output, as rationing induces a decoupling between the two. I suggest an empirical test to estimate the extent of rationing through micro data on sales. Last, I consider the labor wedge as microfounded by rationing, and extend the model to non-homothetic utility with preference for quality.

1 Introduction

Keynes’s (1936) foundational theory of economic fluctuations pinpoints cases where market clearing is held back by insufficient aggregate demand. Similarly, the New Keynesian (NK) framework has inherited the idea that output is demand-determined. Demand acts as the ‘short’ side of the market; suppliers passively accommodate any level of demand; unemployment results from insufficient aggregate demand.

The opposite scenario, where full employment is precluded by *excess* demand met by supply-side rationing, has only found concise analysis in the so-called “Old” Keynesian (OK) model (Barro,

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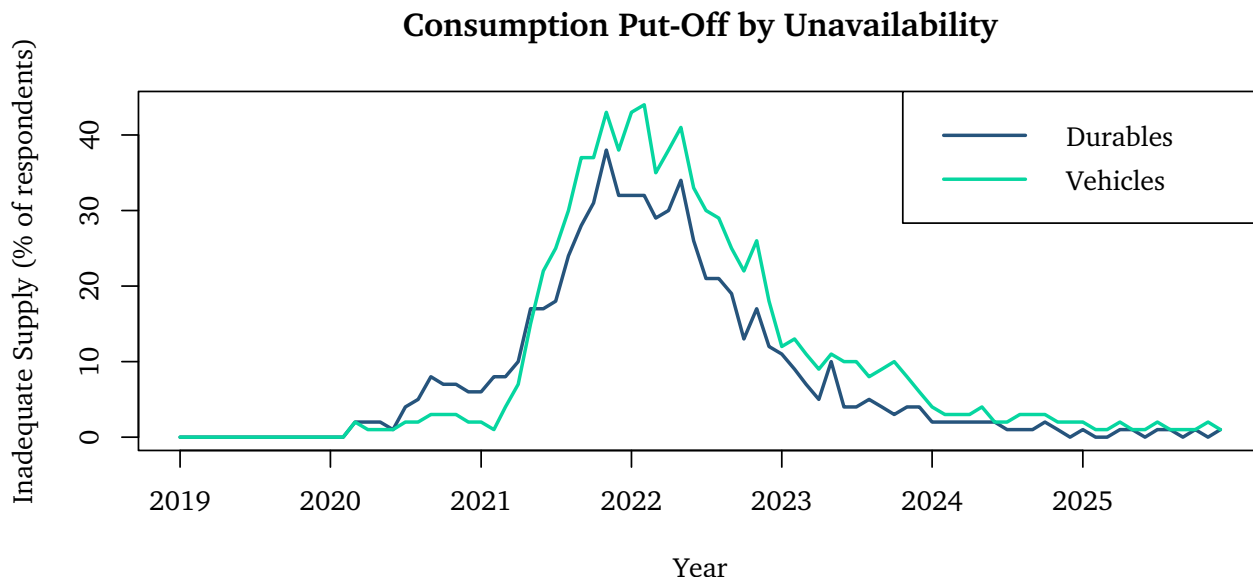


Figure 1 – Consumption postponed due to perceived supply unavailability between 2019 and 2026. The chart illustrates the percentage of respondents reporting inadequate supply for durables (blue) and vehicles (teal). Author’s calculation on data from the University of Michigan, Survey Research Center, Surveys of Consumers

2025; Barro & Grossman, 1971, 1974). This model and its refinements (Michaillat & Saez, 2015) remained a niche in the literature on nominal rigidities, partly due to complex regime-dependent equilibrium conditions, or strong reliance on unobservables. Whether this omission is of practical importance depends on the type of economic phenomena that supply-side rationing can capture relative to traditional Keynesian models (Figure 1).

I contend that rationing can generate interesting phenomena involving markup cycles, state-dependent capacity constraints, and business cycle accounting patterns.

To begin with, nominal rigidities can induce a peculiar form of capacity constraint. Typically, rationed firms are characterized as *profitable* firms, but unable to scale production due to factor inelasticity and thus adjusting prices to address their planning problem. In the presence of nominal rigidities, instead, an alternative definition emerges: *profitability itself* determines productive capacity for fixprice firms. If price adjustments are precluded, productive firms constrain output to mitigate losses, transforming nominal price rigidity into a factual capacity constraint.

Second, rationing can stimulate markups in response to demand shocks through ‘compositional’ dynamics. This result is particularly compelling, as markups *do* rise with demand (Burstein et al., 2025), whereas the standard NK framework typically predicts the opposite (Nekarda & Ramey, 2020). This departure from the established NK literature represents the central contribution of this paper. In the NK model, demand shocks raise employment and lower the marginal productivity of labor; and with sticky prices, this results in an erosion of markups. While this remains true for each individual firm, rationing affects the *composition* of the consumption bundle, reallocating

consumption toward unrationed, high-margin commodities. In the aggregate, then, markups can rise due to a composition effect, even as individual firm markups decline.

From a business cycle perspective, *cyclical* refers to output rather than the underlying structural shocks, and rationing makes the relationship between markups and output nontrivial: price dispersion introduces first-order distortions that decouple ‘physical’ output from welfare-relevant output. As a consequence, measured output might effectively expand while welfare-relevant output declines. Lower welfare-relevant output means that, while the econometrician observes a boom in physical output, the household’s loss of variety raises the marginal utility of consumption sharply and stimulates labor supply despite the apparently low real wages. This phenomenon can account for empirical patterns in markup cyclical, and will be formalized through GDP imputation via ‘statistical’ price indices – such as Laspeyres, Paasche, or Palgrave – and connected to the labor wedge (Chari et al., 2009).

In the remainder of this article, I present a New Keynesian model with endogenous rationing, which I playfully term the ‘New Old’ Keynesian (NOK) model. I discuss the relationship of this paper to the relevant literature in Section 2. Section 3 presents and interprets the model and its key equations, including a discussion of the novel mechanism at play. It also characterizes the dynamics of the model in response to positive shocks to the real interest rate (Section 3.1) and the nominal interest rate (Section 3.2), and reconciles markup cyclical through a “mismeasurement” argument. Section 4 discusses the labor wedge and the extension to non-homothetic utility (Jaimovich et al., 2019). Section 5 concludes the paper.

2 Related Literature

The primary contribution of this paper is to equip the NK model with the capacity to capture procyclical markups in response to demand shocks. This is a phenomenon observed in the data that standard NK models have historically failed to replicate (Nekarda & Ramey, 2020). Recent empirical work by Burstein et al. (2025) provides evidence for compositional dynamics analogous to those generated by my model: using firm-level data, they document that aggregate markup procyclical is, at least in part, driven by a compositional shift in expenditure across varieties. However, Burstein et al. focus on productivity shocks within oligopolistic competition relying on the theoretical equivalence between productivity and demand shocks, which no longer holds under sticky prices and decreasing returns to scale.

I refer to this model as ‘New Old’ Keynesian. Barro (2025) recently revived the debate about rationing or excess demand by promoting the Old Keynesian (OK) model he developed in the 70s (Barro & Grossman, 1971, 1974). His last contribution emphasizes where the OK model stands relative to NK model, and why it is preferable to model nominal rigidities. The OK model takes the concept of equilibrium – defined as the point where demand equates supply – much more seriously than the NK framework, which effectively *assumes* equilibrium by ignoring the supply side altogether. However, it relies on a static “disequilibrium” framework – for tractability –, which

limits its applicability dramatically. It remains also overly naive insofar as a formal price-setting mechanism is absent¹ and its primary channel depends entirely on the direct utility derived from real money balances².

Formalizing their ideas into a New Keynesian DSGE model, I unify the rationing and non-rationing regimes within a single framework that allows for non-Walrasian equilibria, establishing a formal connection to the classical literature on non-Walrasian equilibria (Bénassy, 2005; Malinvaud, 1977).

This effort to revive capacity constraints with nominal rigidities is gaining traction beyond this contribution. Holden (2025) introduces random rationing into a continuous time NK model, showing that when inflation erodes real prices and marginal costs exceed prices, firms optimally choose to stock out rather than satisfy demand at a loss. Holden leverages this friction primarily to reevaluate the welfare costs of trend inflation and to derive a convex, backward-bending Phillips curve. With a similar rationing constraint, I address a different macroeconomic puzzle – that extensive-margin stockouts can lead to procyclical aggregate markups. This forced reallocation is impossible in Holden’s model because rationing is random and uncorrelated with margins, while this is the exact engine of the markup cyclical argument. Holden’s (2025) random rationing model retains beautiful mathematical tractability, but his choice of random rather than sales-capped rationing – where “sellers are capping the quantity they would sell to any individual consumer” (*ivi*, p. 7) – is a trade-off with realism, put forward to avoid “implausibly steep Phillips curves” (*ibidem*). I address this claim in Section 3.3, leveraging the distinction between welfare-relevant and expenditure-weighted variables.

Misallocation connects this paper to the literature on cyclical expenditure shifting across product categories. Jaimovich et al. (2019) document that households systematically “trade down” to lower-priced, lower-quality goods during recessions. Their mechanism is purely demand driven: as aggregate income falls, optimal consumer choice for a non-consumption-homothetic utility function dictates a reduction in both the quantity and the quality of the consumption bundle. In Section 4.2, I incorporate this mechanism into the NOK model, with opposite predictions than Jaimovich et al. (2019).

Last, it is worth mentioning that endogenous rationing can provide a structural, micro-founded explanation for the ‘labor wedge’, the centerpiece of the NK critique formulated by Chari et al. (2009). In neoclassical macroeconomics, fluctuations in this wedge act as a time-varying tax on labor. Because the NK model struggles to generate this friction endogenously, it resorts to reduced-form, theoretically loose ‘(wage) markup shocks’. The rationing framework eliminates this reliance on *ad hoc* shocks, that retain large explanatory power even after corrections (Galí et al., 2011;

¹All three papers (Barro, 2025; Barro & Grossman, 1971, 1974) examine cases where the price level is fixed at a level different from market clearing, and real wages are fixed at a level compatible with market clearing.

²As Barro and Grossman (1974, p. 92) note, “Holdings of money balances provide a medium which ties together the time paths of [notional demand] and [labor supplied]”. Clearly, “the presence of money in the utility function is necessary to obtain an interesting concept of aggregate demand in a static environment because without money, consumers would mechanically spend all their income on the produced good (Say’s law)” (Michaillat & Saez, 2015, p. 523).

Smets & Wouters, 2007). As Section 4.1 highlights, extensive-margin stockouts endogenously drive a wedge between the household's marginal rate of substitution and the measured marginal product of labor. By forcing consumers toward higher-margin varieties, rationing acts as a hidden inflation tax that directly suppresses the welfare-relevant real wage.

3 The Model

The NOK model introduces rationing into an otherwise standard New Keynesian framework (Galí, 2015). Consider an economy populated by a continuum of identical households and a continuum of monopolistically competitive firms $i \in [0, 1]$, each facing a constant probability $1 - \theta$ of resetting its price in any given period. The final good is produced using labor under decreasing returns to scale ($0 < \alpha < 1$) given the aggregate linear technology A_t .

I microfound rationing as a quantity-constrained household optimization problem. The household problem is such that unrationed goods follow a clean Dixit-Stiglitz demand, while forced substitution is taken as given by the households and naturally falls out of the math.

Assume the household values consumption according to the standard CES aggregator, but we partition the unit interval into a set of unrationed goods U_t and a set of rationed goods R_t . For goods $r \in R_t$, the household faces a physical constraint: they can only consume the firm's *maximum profitable supply*, $\mathcal{Y}_t(r)$. The maximum profitable supply is defined at that level of firm output that, at the given nominal wage and final price, attains the minimum markup floor $\underline{\mathcal{M}}$ (for instance, $\underline{\mathcal{M}} \equiv 1$ if firms ration at zero profits). Therefore, the household can treat this quantity as an exogenous endowment. I can rewrite the baseline CES utility aggregator as:

$$C_t = \left(\int_{u \in U_t} C_t(u)^{\frac{\varepsilon-1}{\varepsilon}} + \int_{r \in R_t} \mathcal{Y}_t(r)^{\frac{\varepsilon-1}{\varepsilon}} \right)^{\frac{\varepsilon}{\varepsilon-1}} \quad (1)$$

For instance, with pure-labor DRS technology, $\mathcal{Y}_t(i) = A_t \left(\frac{P_t(i) \alpha A_t}{\underline{\mathcal{M}} W_t} \right)^{\frac{1-\alpha}{\alpha}}$. Let $\mathcal{R}_t$ denote the pure “utility contribution” of the rationed goods, which the household takes as given during their expenditure minimization for a specific period. Similarly, define an unrationed sub-bundle $\mathcal{U}_t$, chosen such that the households satisfy a given level of aggregate consumption C_t , achieving the remaining required utility strictly from the unrationed goods.

$$\mathcal{R}_t \equiv \int_{r \in R_t} \mathcal{Y}_t(r)^{\frac{\varepsilon-1}{\varepsilon}} \quad \mathcal{U}_t \equiv \int_{u \in U_t} C_t(u)^{\frac{\varepsilon-1}{\varepsilon}} \quad (2)$$

which means the aggregate CES relationship equivalently simplifies to:

$$C_t = (\mathcal{U}_t + \mathcal{R}_t)^{\frac{\varepsilon}{\varepsilon-1}} \quad \mathcal{U}_t = C_t^{\frac{\varepsilon-1}{\varepsilon}} - \mathcal{R}_t \quad (3)$$

The household's problem is now a standard conditional expenditure minimization. They seek

to minimize the cost of acquiring the unrationed sub-bundle $\mathcal{U}_t$:

$$\min_{C_t(u) \in U_t} \int_{u \in U_t} P_t(u) C_t(u) \quad \text{subject to} \quad \left(\int_{u \in U_t} C_t(u)^{\frac{\varepsilon-1}{\varepsilon}} \right)^{\frac{\varepsilon}{\varepsilon-1}} \geq \mathcal{U}_t^{\frac{\varepsilon}{\varepsilon-1}}$$

Because this sub-problem is completely free of constraints (by definition, U_t contains only unrationed goods), it yields a clean Dixit-Stiglitz demand function strictly bounded to the unrationed market:

$$C_t(u) = \left(\frac{P_t(u)}{P_t^U} \right)^{-\varepsilon} \mathcal{U}_t^{\frac{\varepsilon}{\varepsilon-1}}$$

where the unrationed price index is exactly what one would expect:

$$P_t^U = \left(\int_{u \in U_t} P_t(u)^{1-\varepsilon} \right)^{\frac{1}{1-\varepsilon}}$$

To compute the aggregate price level under this split-integral regime, note that there is no longer a single, dual price index that satisfies both total expenditure and the marginal cost of utility. The household is physically constrained in the rationed sector, and goods in this sector are infra-marginal (the household wants to consume more of them at their posted prices but cannot). Therefore, any marginal change in aggregate consumption, labor supply, or savings is driven entirely by the unrationed sector, although the size of this sector and its complement are state-dependent.

To compute the marginal cost of aggregate utility, I start with the split CES aggregator established in 1. Total expenditure E_t is split into involuntary (predetermined) expenditure on rationed goods and discretionary expenditure on unrationed goods. Because $\mathcal{U}_t$ is defined as the utility sum $\int_{u \in U_t} C_t(u)^{\frac{\varepsilon-1}{\varepsilon}}$, the actual composite quantity of unrationed goods is $\mathcal{U}_t^{\frac{\varepsilon}{\varepsilon-1}}$. The welfare-relevant price level can be equivalently written as:

$$\tilde{P}_t = P_t^U \mathcal{U}_t^{\frac{1}{\varepsilon-1}} C_t^{-\frac{1}{\varepsilon}} \quad \tilde{P}_t = P_t^U \left(1 - \frac{\mathcal{R}_t}{C_t^{\frac{\varepsilon-1}{\varepsilon}}} \right)^{\frac{1}{\varepsilon-1}} \quad (4)$$

which also allows to rewrite the CES demand for unrationed as:

$$C_t(u) = \left(\frac{P_t(u)}{\tilde{P}_t} \right)^{-\varepsilon} C_t \quad (5)$$

The existence of a welfare-relevant price index distinct from the CES price aggregator is a known result under rationing (Holden, 2025), and will be crucial for exploring the cyclical properties of markups and wages entailed by the model (see Section 3.2). To understand how the different price levels relate, I attach a summary illustration of the price patterns generated by a rationing economy in response to a demand shock in Figure 2. This is an anticipation of Figure 4, as it shows the cross-sectional properties of prices and price levels corresponding to that simulation.

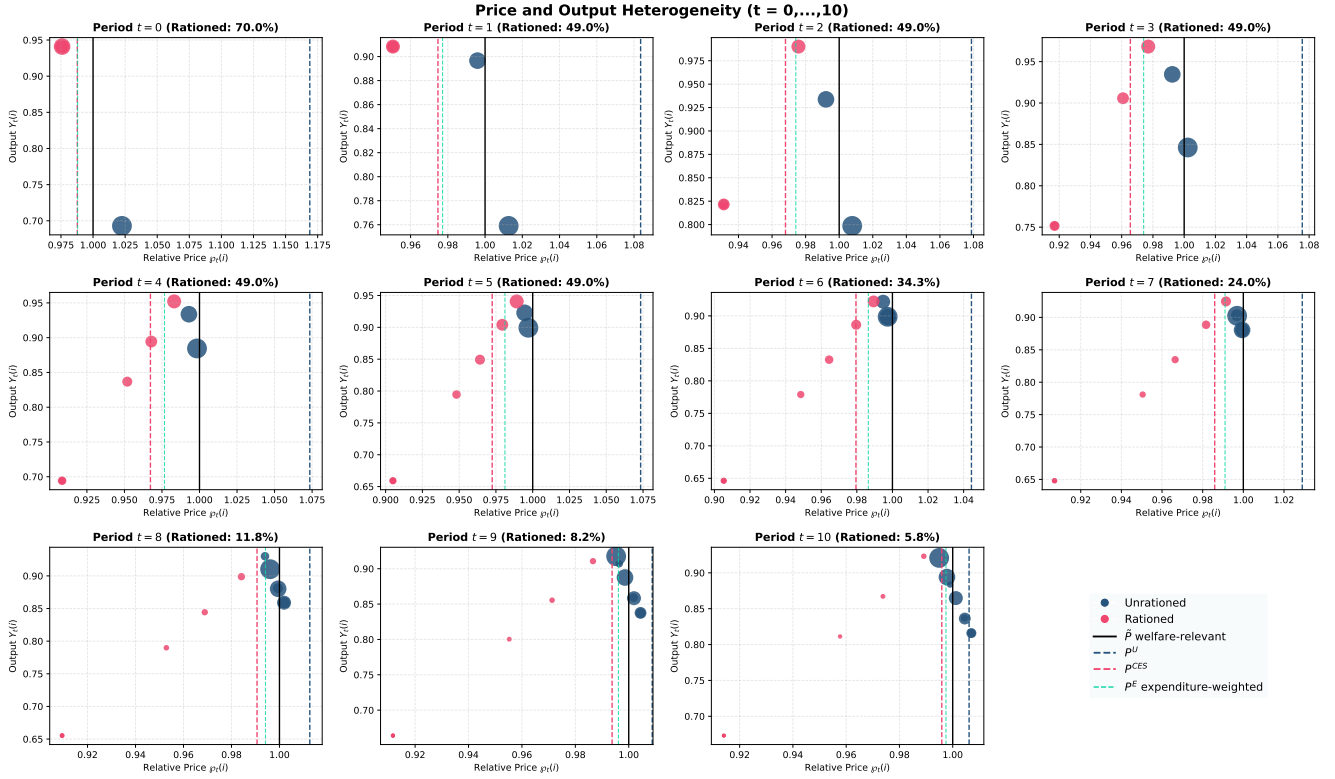


Figure 2 – Scatter plot of price and output heterogeneity in Figure 4 for $t = 0, \dots, 10$. Bubbles represent individual firms, classified by rationing status (color) and Calvo mass (size). Vertical bars locate the prevailing price indices.

This illustration warrants two observations concerning individual and aggregate prices.

First, the rationing mechanism reveals a distinctive pattern. For unrated goods, the standard microeconomic inverse relationship between prices and quantities prevails. Conversely, for rationed goods, the opposite holds: as price becomes the capacity bottleneck, higher prices correspond to higher demand. An implication of this phenomenon is that the extent of rationing resulting from nominal rigidities can be tested empirically by examining the correlation between quantities and prices. Naturally, this test faces several limitations: it is invalidated if goods are not perfect substitutes – for instance if prices reflect unobservable quality differences, as in Jaimovich et al. (2019) – and it requires that the capacity constraint be attributable solely to price rigidities. Because other sources of rationing, where firms respond to constraints by raising prices (Burstein et al., 2025), may imply opposite dynamics, this empirical test may still offer a potential diagnostic tool for identifying the source of rationing and disambiguating between competing mechanisms.

The second observation concerns the aggregate price level. The illustration confirms a clear ordering between the various price indices, namely $P_t^{\text{CES}} \leq \tilde{P}_t \leq \tilde{P}_t \leq P_t^U$, where $\tilde{P}_t$ denotes the expenditure-weighted price level (detailed in Section 3.2). This ordering arises intuitively from the definition of the indices, and the derivation is omitted here for brevity. However, the fact that the expenditure-weighted price is systematically lower than the welfare-relevant prices will be key to address the markup cyclicity puzzle.

The welfare-relevant price $\tilde{P}_t$ is the shadow price of consumption that households use when

evaluating their marginal rate of substitution. The household equates the marginal disutility of labor to the marginal utility of consumption, scaled by the real wage. Because marginal consumption happens at $\tilde{P}_t$, the correct deflator is not a standard price index, but the shadow price.

$$\frac{W_t}{\tilde{P}_t} = \frac{N_t^\varphi}{C_t^{-\sigma} - \xi_t} \quad (6)$$

$$C_t^{-\sigma} - \xi_t = \mathbb{E}_t \left[\frac{\beta I_t}{\tilde{\Pi}_{t+1}} (C_{t+1}^{-\sigma} - \xi_{t+1}) \right] \quad (7)$$

where I_t denotes the gross nominal interest rate and ξ_t is the Lagrange multiplier associated with the economy-wide capacity constraint $\mathcal{D}_t \leq \mathcal{Y}_t$ (see Theorem 3.3). Standard NK models always enforce *Walrasian* equilibria: the aggregate capacity constraint is always slack and the multiplier disappears. When notional and effective demand (supply) are decoupled, the multiplier becomes a crucial determinant of the household's Euler equation, as individual optimality is still satisfied despite the household fails to equalize marginal utility of consumption across periods.

Standard general equilibrium arguments ensure that an equilibrium may exist, provided that the price index varies continuously when firm cohorts transition from unrationed to rationed regimes. Continuity is established by the following proposition.

Proposition 3.1 (Global Continuity of the Welfare-Relevant Price). *The welfare-relevant shadow price index $\tilde{P}_t$ is globally continuous.*

Proof. While the expenditure function is trivially differentiable everywhere, the global continuity of its derivative is proved in the Appendix (p. 23). $\square$

Another important implication of this model is that price distortion $\Delta_t = \int_0^1 P_t(i)/\tilde{P}_t di$ exhibits first order effects on output, as the following Proposition shows. This result is foundational to the business accounting remarks that will follow.

Proposition 3.2 (First-Order Cost of Price Dispersion). *Whenever a positive mass of goods is rationed, an increase in price dispersion reduces aggregate CES output at a first-order rate. Moreover, the linear sum of goods and the CES aggregate differ by a term that is first-order in the extent of rationing.*

Proof. See Appendix (p. 26). $\square$

To make the proposition intuitive, consider a mean-preserving spread of the relative prices $\wp_t(i) \equiv P_t(i)/\tilde{P}_t$ that holds the shadow price identity constant. Such a spread lowers some $\wp_t(j)$, making some j goods cheaper in relative terms, and raises others to compensate. The cheaper goods face higher notional demand. If they already lie in R_t or are pushed into R_t by the spread, their actual consumption is capped at the maximum supply, and does not increase in proportion. Hence, the integral $\int_{r \in R_t} \mathcal{Y}_t(r) - \wp_t(r)^{-\varepsilon} Y_t$ becomes more negative. Because this integral is of order $|R_t|$, its

derivative with respect to the price spread is non-zero and negative. Consequently, $\frac{\partial Y_t}{\partial \Delta_t} < 0$ and the effect is truly first-order.

The first-order effect of price dispersion is a key result for the measured cyclicalities of output, and thus for the markup cycles outlined in Section 3.2. Indeed, together with the decoupling of the welfare-relevant price from other price indices, it offers two powerful channels through which econometric accounting and welfare-relevant quantities diverge. Consider the case of an econometrician measuring output and inflation. The econometrician observes total expenditure and expenditure shares. Mismeasurement can simultaneously arise from two distinct channels:

- The econometrician might utilize an empirical price index, such as an expenditure-weighted or a standard CES aggregator, rather than the true welfare-relevant price.
- The econometrician might aggregate sectoral quantities linearly rather than through the welfare-relevant CES aggregator. Under rationing, this discrepancy becomes first-order.

To resonate with the three-equation structure, the model closes for a New Keynesian Phillips Curve (NKPC) and a law of motion for the nominal interest rate such as a Taylor rule. The former depends on the pricing strategy of the firm, that is captured by the following firm optimality condition. The optimal pricing strategy is:

$$P_t^* = \frac{\varepsilon}{\varepsilon - 1} \frac{\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} \mathbb{1}_{t+k}^U Y_{t+k|t} MC_{t+k|t}^U}{\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} \mathbb{1}_{t+k}^U Y_{t+k|t} - \left(\frac{\mathcal{W}}{\varepsilon-1}\right) \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} \mathbb{1}_{t+k}^R \mathcal{Y}_{t+k|t}} \quad (8)$$

where $\mathcal{W} \equiv \left[1 + \frac{\alpha}{1-\alpha} \left(\frac{\underline{M}-1}{\underline{M}}\right)\right] \geq 1$ is referred to as the *marginal capacity value of price*.

The equation is similar to the standard pricing function in the textbook NK models, apart from the sum over future levels of rationed output weighted by the marginal capacity value of price. A smaller denominator means that the model features an upward ‘precautionary’ markup, driven by the intensive margin inside the rationed regime. Because $\mathcal{W} \geq 1$ (or exactly one at zero marginal profits), $\varepsilon > 1$, and future rationed output are all strictly positive, this entire term reduces the denominator relative to a standard setup, forcing the reset price P_t^* to be strictly higher.

The logic captured by $\mathcal{W}$ might be summarized as follows: The firm raises its price today not to avoid the *probability* of hitting the capacity wall (firms are indifferent between rationing and unrationing at the margin), but to loosen the severity of the physical constraint *after* it hits the wall (they might get stuck with smaller capped output if their prices are too low). Moreover, a higher P_t^* today means a higher nominal marginal cost is permitted before the $\underline{M}$ floor binds tomorrow, allowing the firm to capture more of the demand despite being rationed.

Before moving on to establish some key properties of the model, I should emphasize that the capacity limits induced by price rigidities not only apply at the firm level, but also impose a hard economy-wide capacity limit. As a consequence, a model with rationing may fail to clear if large demand-side shocks are incompatible with supply-side profitability.

Theorem 3.3 (Non-Walrasian Equilibria). *Any Walrasian equilibrium must satisfy $C_t \leq Y_t^F < \infty$. For a sufficiently large demand shock, only non-Walrasian equilibria exist.*

Proof. See Appendix (p. 28). An economy-wide capacity bound $Y_t^{\mathcal{F}}(V_t)$ binds for a given real wage V_t because a high real wage compresses the capacity of (rationed) low-price firms while the Euler equation requires consumption and real wages to be high when the real rate is low. $\square$

Non-Walrasian equilibria require all firms in the economy to hit their maximum profitable supply and suspend sales. This includes the current mass of price adjusters. If the optimization problem were static, adjusters would never optimally choose a self-rationing price: any price increase up to their break-even point would raise revenues without crowding out demand. However, an adjuster might optimally set a price below their rationing cutoff if the probability of crowding out future demand due to a high reset price is sufficiently large. This allows to derive a necessary condition for non-Walrasian regimes.

Proposition 3.4 (Necessary Condition for Self-Rationing). *Let the current period be $t = 0$ with a demand state such that at any $P \geq P_0$ the firm is unrationed. The firm optimally chooses a price in the rationed region ($P_0^* < P_0$) if and only if:*

$$\mathcal{WY}_0(P) > -\mathbb{E}_0 \sum_{k=1}^{\infty} (\beta\theta)^k \Lambda \left[\mathbb{1}^U \frac{\partial \varpi^U(P)}{\partial P} + \mathbb{1}^R \mathcal{WY}_k(P) \right] \quad \text{for } P = P_0 \quad (9)$$

That is, the marginal gain from raising price out of rationing today is smaller than the marginal loss from applying that higher price in all future states.

In general, this can occur through high price stickiness θ , a large transitory shock. With a sufficiently strong endogenous monetary response, the numerical equilibria always remain market-clearing. However, when the system is shocked by a sufficiently large real rate shock, a non-Walrasian equilibrium can prevail on impact (see Figure 3).

The next two sections examine the dynamics of the model from a theoretical and business cycle accounting perspective. To begin with, I illustrate the fundamentals of the model in Section 3.1. Then, I define a standard³ Taylor rule and illustrate the dynamics of the three-equation model with endogenous monetary policy in Section 3.2. This section includes key implications for markup cyclical. I conclude discussing the foundation of the labor wedge in a model with rationing and with quality in the utility function (Section 4.1).

Impulse Response Functions

The next two sections present the dynamic response of the economy to an exogenous shock under perfect foresight. In both cases, I solve the full non-linear model, using deterministic root-finding for the real interest rate shock and a deep neural network policy function for the nominal interest rate shock.

³The Taylor rule is not optimal. Optimal monetary policy is beyond the current scope and is postponed to further research.

In both instances, the economy is represented by a state vector of size $P + T + 1$, where P denotes the subset of parameters allowed as inputs, and T denotes the length of the roll-out period tracking each Calvo cohort's mass and price, plus the exogenous shock. To interpret the remainder of the IRFs, note that shocking a symmetric equilibrium with one Calvo cohort adjustment at a time implies that, at any period t , only $t + 1$ heterogeneous prices exist. A natural consequence is that, on impact, only two masses of internally homogeneous firms exist, of size θ and $1 - \theta$ respectively, meaning that either block can be rationed or unrationed.

The calibration utilized in the simulation is summarized in Table 1.

Table 1 – Parametric calibration in the simulated economy.

Parameter	Description	Value
β	Discount factor	0.99
σ	CRRRA parameter	1.0
φ	Inverse Frisch elasticity	1.0
ρ	Persistence of shock	0.75
A	Productivity	1.0
α	Decreasing returns to labor	0.80
θ	Calvo price-stickiness	0.70
ε	Elasticity of substitution	10.0
$\underline{\mathcal{M}}$	Markup floor (fraction of $\mathcal{M}$)	0.98

3.1 A Real Interest Rate Shock

In this section, I perturb a zero-inflation symmetric steady state with an AR(1) shock to the real interest rate to generate the corresponding perfect-foresight impulse response functions. The model can be easily solved globally via Newton–Raphson root-finding. Indeed, the time- t real interest rate in a perfect-foresight framework is exactly $\frac{\beta I_t}{\Pi_{t+1}}$, which is set equal to one in the terminal condition T . Given an exogenous shock ν with autoregressive parameter ρ , the entire path of the real rate is known for every period t as $1 + \rho^t \nu$. From the household's first-order condition (FOC), it follows that $\frac{\lambda_t}{\lambda_{t+1}} = \frac{\beta I_t}{\Pi_{t+1}} = 1 + \rho^t \nu$. Once the entire sequence $\left\{ \frac{\lambda_t}{\lambda_{t+1}} \right\}_0^{T-1}$ is determined, the remaining static equilibrium conditions can be solved sequentially. The results are shown in Figure 3.

The combination of a high margin floor ($\underline{\mathcal{M}} = 1.08$), aggressive DRS, and high final-good elasticity of demand is such that a 0.5% shock to the real rate suffices to push the economy into a non-Walrasian equilibrium on impact. The excess demand that the economy cannot absorb is captured by an instantaneous, isolated spike in the multiplier ξ_t . An implication of this jump in the excess demand multiplier is that households accept a real wage below their marginal rate of substitution, with the unique consequence of increasing output efficiency: while rationing typically increases misallocation – keeping output below steady-state levels in a Walrasian equilibrium – a decoupling of the wage from the MRS allows for a temporary improvement in productive efficiency. This anomalous result disappears as soon as the economy returns to a market-clearing state.

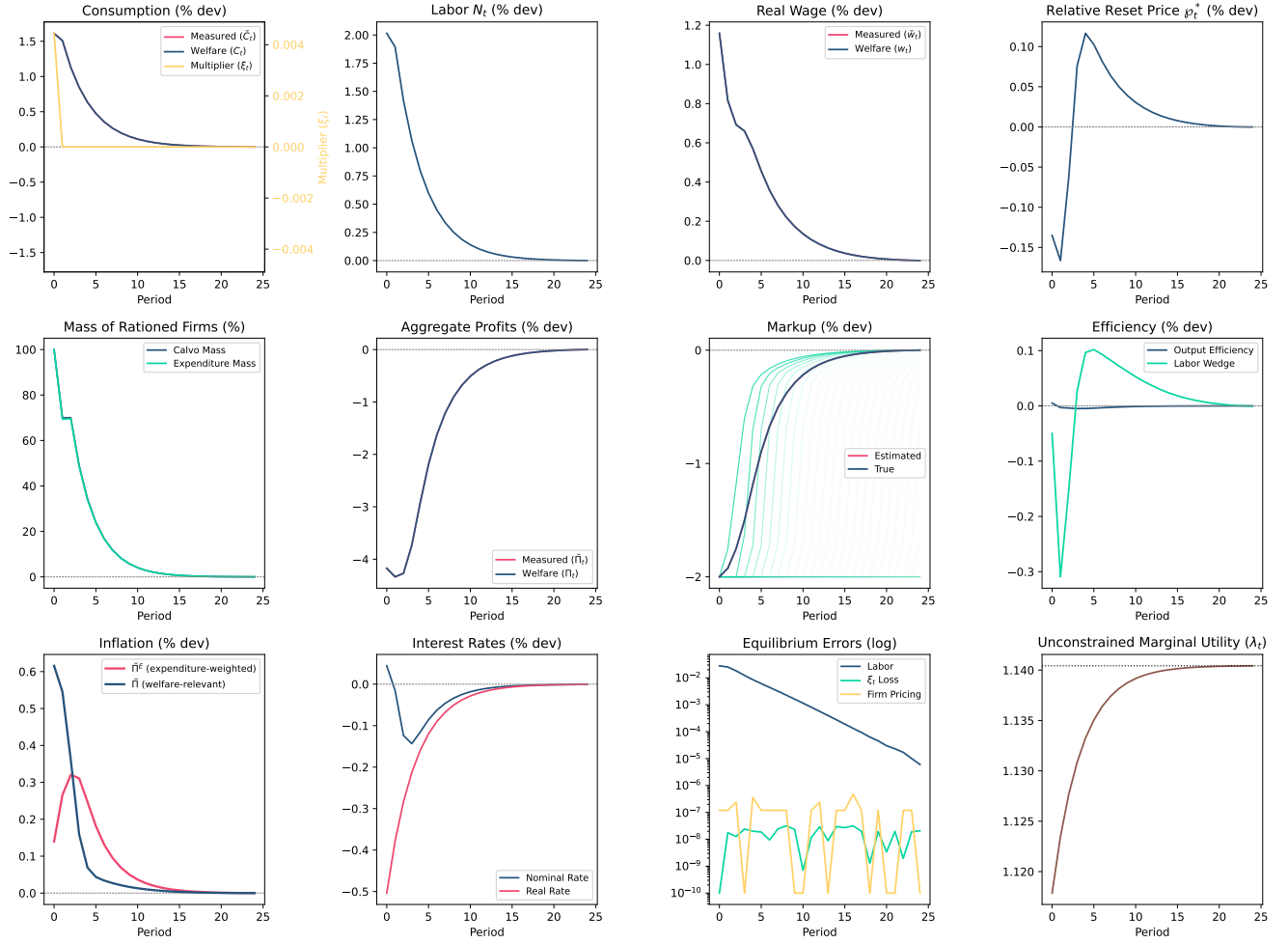


Figure 3 – IRFs to an AR(1) exogenous shock to the real interest rate (0.5%). Blue lines indicate welfare-relevant quantities and teal lines denote cohort-specific markups. The last subplot report the (log) equilibrium errors realized by the root-finding algorithm and measures accuracy.

Interestingly, the optimal pricing response with perfect-foresight consists in a self-rationing adjustment on impact. The real rate shock creates a transitory demand boom that firms decline to fully accommodate because raising prices enough to clear their market would leave them uncompetitively expensive for approximately the 2 or more quarters after demand normalizes – and with Calvo probability 0.70, they expect to be stuck with that price through the normalization.

3.2 A Nominal Interest Rate Shock

In this section, I present the perfect-foresight IRFs in response to an AR(1) 1% exogenous shock to the monetary policy rule.

The policy functions are approximated using a deep neural network that maps the state vector consisting of $K - 1$ past relative log prices, the monetary policy shock, and the four structural parameters α , θ , ϵ , and markup floor percentage⁴ to aggregate consumption C_t , the normalized nominal wage, and the normalized optimal reset price. In this specification, I omit the multiplier ξ_t associated with the economy-wide capacity constraint, as the endogenous monetary policy rule prevents current demand from rationing the entire economy due to the sharp inflationary response this entails.

In contrast to Figure 3, I report not only the welfare-relevant paths but also the empirical paths measured by an econometrician following standard macroeconomic accounting practices. Suppose the econometrician does not observe physical output or the welfare-relevant price index. Instead, they observe total expenditure and expenditure weights in the consumption basket. Rather than comparing the model against the unconstrained CES price index, I assume the econometrician infers total output by deflating expenditure using an *expenditure-weighted* price index. In our model, this is defined as follows: denote $Y_t(j)$ as the realized output of a cohort that last reset its price in $t - j$. The expenditure-weighted price index is given by:

$$\check{P}_t \equiv \sum_{j=0}^{\infty} \omega_t(j) P_t(j) \quad (10)$$

where $\omega_t(j) \equiv (1 - \theta)\theta^j$ denotes the Calvo mass. If the econometrician deflates expenditure using this index, measured GDP diverges from welfare-relevant output by the ratio of the price levels:

$$\check{C}_t = \frac{E_t}{\check{P}_t} = C_t \left(\frac{\tilde{P}_t}{\check{P}_t} \right) \quad (11)$$

The $\check{P}$ -price level is the model equivalent of a Palgrave price index⁵, and resonates with actual accounting practices while remaining relatively stylized. On the one hand, it captures the fact that the price level is imputed in actual practice by attaching weights to the single varieties, composing

⁴Rather than hard-coding the margin floor, I define $\underline{M}$ as the *percentage of steady state markups* that firms are willing to internalize, for computational generalizability.

⁵The Palgrave price index is a weighted arithmetic mean of price relatives, where the weights are the comparison period value shares. For this or other definitions, refer to Balk (2008).

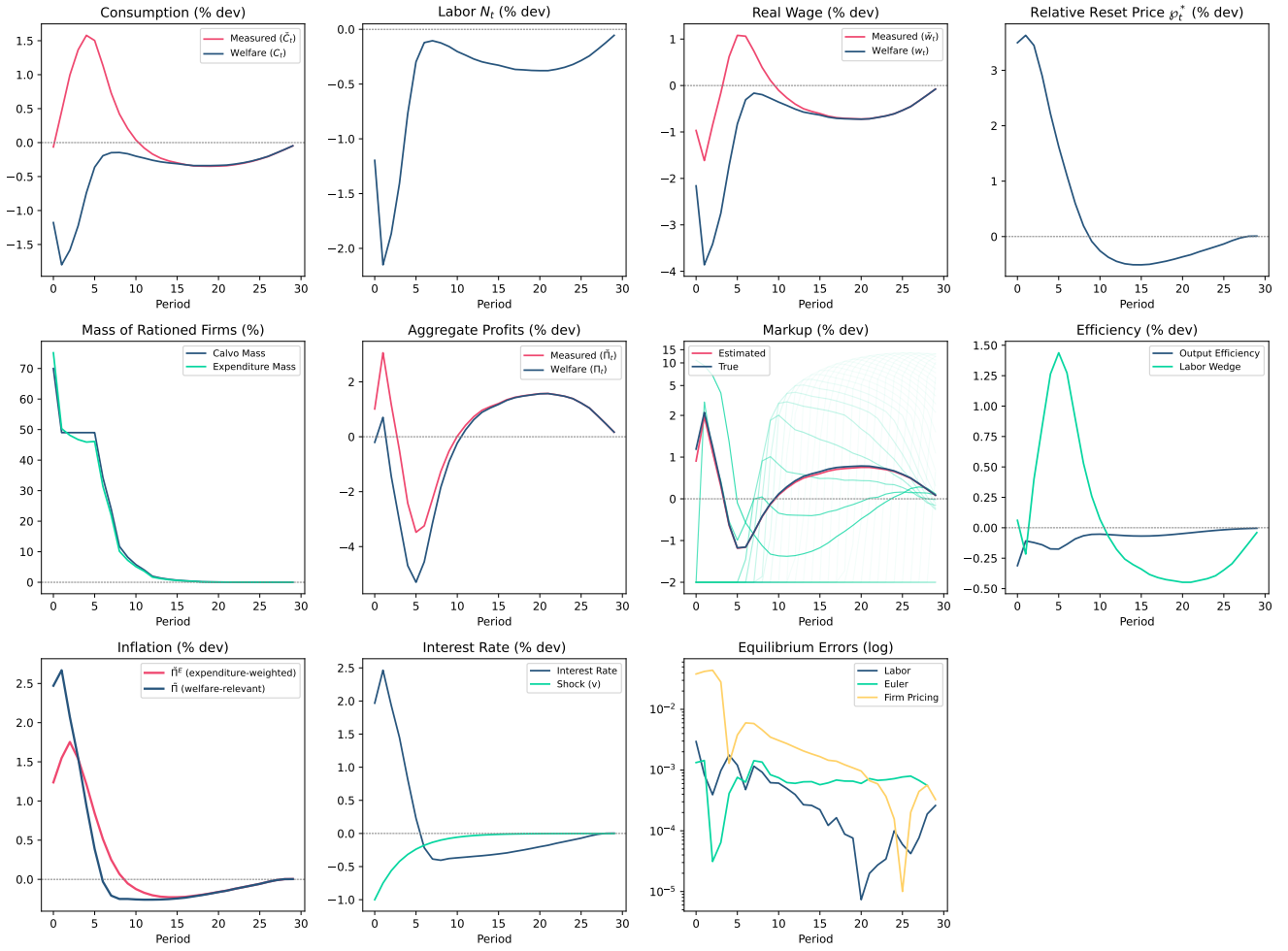


Figure 4 – IRFs to an AR(1) exogenous monetary policy shock (1%). Blue lines indicate welfare-relevant quantities, pink lines indicate the corresponding expenditure-weighted measures, and teal lines denote cohort-specific markups. The last subplot report the (log) equilibrium errors realized by the network and measures accuracy.

a time-varying consumption basket. On the other hand, it is a simplifies measure relative to Fisher or Törnqvist, as it does not take a combination over multiple periods. Moreover, it assumes that the econometrician does not observe or estimate output from the supply side directly.

I also define measured markup as the inverse of the labor share, in accordance with the estimations performed by Nekarda and Ramey (2020). In the original model, the ‘true’ aggregate markup is expenditure-weighted, that is calculated as the arithmetic average of individual firm markups weighted by their nominal expenditure shares, $\mathcal{M}_t = \sum_{j=0}^{\infty} \omega_t(j) \mathcal{M}_t(j)$. Because an individual firm’s markup is simply its nominal price divided by its nominal marginal cost, weighting these individual markups by their nominal expenditure shares is tracking the exact same object as the inverse of the labor share.

The disconnect between welfare-relevant output and measured output mirrors the result in Holden (2025). However, its implication for markup cyclicity is critical and cannot be reproduced by his random rationing model. These are shown in Figure 4.

When the positive demand shock hits the economy, the extensive margin reallocation of

expenditure from low-margin rationed varieties to high-margin unrationed varieties drives a significant expansion in the aggregate markup. Because the labor-share proxy relies entirely on aggregate nominal flows, the choice of the price index does not affect the empirical estimates of the aggregate markup, and the econometrician captures this shift accurately. The measurement illusion, however, arises on the quantity side. While true household welfare and physical consumption contract due to the shadow inflation tax imposed by quantity constraints, the current-weighted empirical price index understates this utility penalty. Deflating nominal expenditure by the naive $\tilde{P}_t$ aggregator mathematically generates a phantom boom in measured consumption. Consequently, the simultaneous expansion of measured consumption and measured markups observed in the data does not reflect a generalized economic expansion, but rather masks a supply-side bottleneck and a contraction in true economic welfare. This raises the possibility that the markup procyclicality documented by Nekarda and Ramey (2020) may be a measurement artifact, that can be uncovered once the NK model is equipped with a supply-side mechanism for the determination of output.

The mechanism is summarized as follows: when unprofitable firms ration demand, expenditure is forcibly reallocated toward unrationed, high-margin varieties. This process raises markups, reduces welfare, and exacerbates misallocation, ultimately suppressing output. However, the decoupling of the welfare-relevant price index from standard empirical price aggregators during rationing induces significant measurement error when inferring output from total expenditure. Specifically, if a Laspeyres-type price index – such as the Palgrave index – is employed, the decline in welfare-relevant output can be mismeasured as an increase. Consequently, markups appear countercyclical with respect to welfare-output – consistent with standard NK theory – but procyclical with respect to measured output. This divergence provides a novel resolution to the markup puzzle identified by Nekarda and Ramey (2020) and represents the primary contribution of this paper.

3.3 The New Keynesian Phillips Curve

In his relevant working paper, Holden (2025) argues that models with deterministic quantity-capped rationing generate an implausibly steep NKPC and thus opts for random rationing (via idiosyncratic demand shocks), which effectively results in a flatter slope. This section shows the NKPC implied by the NOK model and addresses that critique directly. This NKPC is illustrated in Figure 5. The phase plots map current inflation against the output gap for contractionary and expansionary shocks. However, rationing is an *asymmetric* shock, that hits during monetary expansions rather than contractions. Under contractions, then, both the welfare and measured plots behave identically – as the output gap turns negative, inflation falls mildly. This creates a standard and relatively flat linear slope consistent with the textbook predictions where nominal rigidities dominate.

Under expansions the plots diverge significantly. The measured curve is L-shaped and convex: as the measured output gap exceeds 1%, the curve bends upward and inflation accelerates rapidly with very little additional gain in measured output. The L-shape remains sharp, although the top of the curve exhibits a ‘rightward loop’ because it takes time for measured inflation to

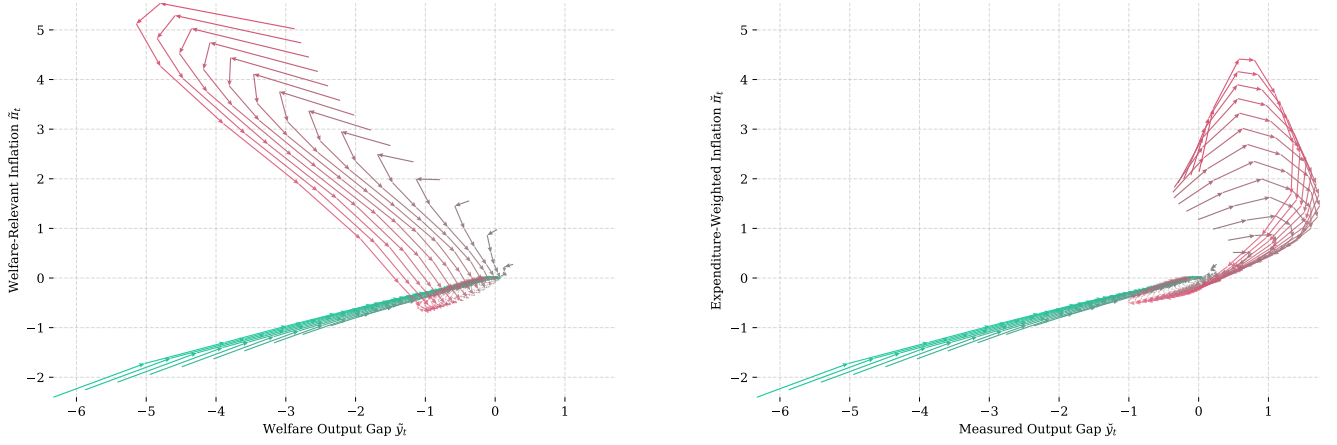


Figure 5 – NKPC implied by 30 simulated economies initialized across a grid of shock sizes ranging from -2.5 to $+2.5$ standard deviations. All variables are expressed as percentage deviations from their steady-state values. The colormap represents the initial shock size (teal for tightening, red for loosening). Path trajectories fade over time. The left-column plots welfare-relevant inflation, the right-panel plots expenditure-weighted inflation.

adjust, given that forced reallocation is accounted as a reweighting of expenditure and fails to capture the underlying spike in the costs of living. In contrast, the welfare-relevant curve is clearly backward-bending, because the welfare-relevant output gap never expands – at the onset of the expansionary shock where measured inflation peaks, the welfare output gap instantaneously hooks backward. As the shock plays out, the welfare gap drops even further into negative territory.

Summing up, the implausible steepness warned by Holden is effectively reproduced by the measured-NKPC’s L-shape. However, the backward-bending welfare-NKPC remains consistent with his findings. This duality is potentially relevant for modern macroeconomic empirical debates about the steepness of the Phillips curve, although a more detailed analysis is deferred to further research.

4 Extensions

4.1 Labor Wedge

As rationing affects marginal utility, either by introducing a non-Walrasian wedge or by introducing first-order losses from price dispersion, its consequences naturally carry over to the MRS and thus the labor wedge. In business cycle accounting, the labor wedge is defined as a wedge between the real wage and the marginal rate of substitution:

$$1 - \tau_{lt} = \frac{U_{lt}/U_{ct}}{A_t F_{lt}} = \frac{MRS_{c,l}}{MPL_t} \quad (12)$$

In their influential critique of the NK model, Chari et al. (2009) argue that quantitative NK models rely on non-structural shocks to match aggregate data, rendering them unsuitable for policy analysis. In particular, they contend that the wage markup shock mechanically acts as

a reduced-form labor wedge that distorts the static relationship between the marginal rate of substitution of consumption for labor and the marginal product of labor. This reduced-form wedge is observationally equivalent to distinct structural frameworks with conflicting policy implications: for instance, if the wedge represents fluctuations in union bargaining power, the optimal policy is to restrict union monopoly power, and if the wedge reflects exogenous preference shocks, the equilibrium allocations are efficient and *laissez-faire* is optimal. Aggregate data cannot distinguish between these structural interpretations, and the shock cannot be considered invariant to policy interventions.

The NOK model naturally endogenizes the fluctuations of the labor wedge by explicitly separating the true rationing effect from the output mismeasurement effect. The true movement of the labor wedge is driven strictly by capacity rationing. The forced substitution and variety loss generated by rationing lead to a spike in the welfare-relevant shadow price, which causes true household consumption to contract. Because the marginal utility of consumption rises while physical labor remains constrained, the true marginal rate of substitution is suppressed, driving up the labor wedge endogenously.

On top of the endogenous labor wedge generated by rationing, the output measurement error would also affect the business cycle accounting exercise carried out by Chari et al. (2009). The measurement problem operates exclusively on the *quantity side*, to create the illusion of an exogenous shock via the usual argument: The econometrician deflates nominal expenditure with an empirical price index that ignores physical quantity constraints, and this mismeasurement produces a phantom expansion in measured consumption. According to this narrative, when standard NK accounting observes a boom in consumption occurring simultaneously with a collapsing labor wedge, it incorrectly attributes the data to an exogenous shock to worker preferences or bargaining power.

4.2 Non-Homothetic Utility

In this section, I extend the rationing model to an environment where production occurs along both quantity and quality margins, and where households exhibit non-homothetic preferences. This mechanism was employed by Jaimovich et al. (2019) to rationalize consumption reallocation along the quality margin during booms, under the assumption that higher prices serve as a proxy for quality. In their framework, as consumption quantity increases, the household's marginal utility of quality rises; since quality-intensive varieties are more labor-intensive, this shift allows for an increase in aggregate labor supply for a given real wage. As this mechanism relates to some core features of the NOK model – namely, the labor wedge and expenditure reallocation – I consider it as a natural extension for this analysis.

Firms produce goods using the following technology:

$$Y_t(i) = \frac{A_t}{Q(Q_t)} N_t(i)^\alpha \quad (13)$$

where $Q(Q_t) \geq 1$ is an increasing function representing the *labor-intensity penalty for producing*

quality. Nominal costs $TC(Y_t(i))$ become:

$$C(Y_t(i)) = W_t N_t(i) = W_t \left(\frac{Y_t(i) Q(Q_t)}{A_t} \right)^{\frac{1}{\alpha}} \quad (14)$$

The firm produces up to a maximum profitable supply where $C'(\mathcal{Y}_t(i)) = P_t(i)$. That is:

$$\begin{aligned} P_t(i) &= \frac{W_t Q(Q_t)}{\alpha A_t} \left(\frac{\mathcal{Y}_t(i) Q(Q_t)}{A_t} \right)^{\frac{1-\alpha}{\alpha}} \\ \mathcal{Y}_t(i) &= \left(\frac{A_t}{Q(Q_t)} \right)^{\frac{1}{1-\alpha}} \left(\frac{\alpha P_t(i)}{W_t} \right)^{\frac{\alpha}{1-\alpha}} \end{aligned} \quad (15)$$

This equation suggests that positive technological shocks provide a convex capacity expansion. The effect of quality depends on the functional form of Q ; for linear cost functions, this also acts as a convex constraint.

Households maximize non-homothetic utility:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{Q_t^{1-\theta}}{1-\theta} \log(C_t) - \frac{N_t^{1+\varphi}}{1+\varphi} \right] \quad (16)$$

subject to the budget constraint. As households choose C_t, Q_t, N_t to maximize utility, quality Q_t is implicitly influenced by the firm's production decisions through $Q(Q_t)$, which affects prices. With rationing, this problem becomes specially interesting. See this by setting up the split-CES aggregator from Equation 1:

$$C_t = \left(\int_{u \in U_t} C_t(u)^{\frac{\varepsilon-1}{\varepsilon}} + \int_{r \in R_t} \left[\left(\frac{A_t}{Q(Q_t)} \right)^{\frac{1}{1-\alpha}} \left(\frac{\alpha P_t(i)}{W_t} \right)^{\frac{\alpha}{1-\alpha}} \right]^{\frac{\varepsilon-1}{\varepsilon}} \right)^{\frac{\varepsilon}{\varepsilon-1}} \quad (17)$$

Put simply, households can effectively relax the supply bottleneck by “giving up” quality when rationing hits. The utility benefit of higher quality should equalize the utility cost of raising marginal costs and reducing the maximum profitable supply, which forcibly reallocates consumption toward unrationed varieties.

Quality choice introduces a new margin through which the household's optimization affects the effective consumption aggregate, even though the household treats the aggregate structure (the rationing threshold, the price index $\tilde{P}_t$) as given. This is analogous to how the household in Jaimovich et al. (2019) is not a price-taker in the quality dimension: it chooses Q_t to maximize utility given the quality-price schedule, and that choice feeds back into the equilibrium price level. By moving the rationing threshold, then, household they can affect the $\mathcal{R}_t$ partition, which is a key determinant of the welfare-relevant price.

I conclude this extension by deriving one key equation. For simplicity, assume prices are distributed according to the fully differentiable density function f , with the rationing threshold sitting at $\underline{f}$.

The derivative of $C_t^{\frac{\varepsilon-1}{\varepsilon}}$ with respect to quality simplifies to:

$$\frac{\partial C_t^{\frac{\varepsilon-1}{\varepsilon}}}{\partial Q} = -\frac{\varepsilon-1}{\alpha\varepsilon} \left(\frac{A}{Q(Q)} \right)^{\frac{\varepsilon-1}{\alpha\varepsilon}} \frac{Q'(Q)}{Q(Q)} \left(\frac{\alpha}{\underline{MW}} \right)^{\frac{(1-\alpha)(\varepsilon-1)}{\alpha\varepsilon}} \mathbb{E}_{f(r)|_0^f} [P(r)^{\frac{(1-\alpha)(\varepsilon-1)}{\alpha\varepsilon}}] < 0 \quad (18)$$

where $\frac{Q'(Q)}{Q(Q)}$ is the quality semi-elasticity of production costs.

Rationing inverts the central mechanism of Jaimovich et al. (2019). In their framework, recessions induce *trading down* – households substitute toward lower-quality goods because quality is a normal good and income falls. In my model, the opposite occurs: (demand-driven) expansions create the incentive to trade down. When aggregate demand is high, the rationing bottleneck binds tighter and the household faces a hard capacity constraint on total consumption. Because $Q(Q_t) \geq 1$ is increasing in quality, producing higher quality reduces a firm's maximum profitable supply. The household, by accepting lower quality, effectively expands the feasible supply of rationed goods captured by the term $\left(\frac{A_t}{Q(Q_t)} \right)^{\frac{\varepsilon-1}{\alpha\varepsilon}}$ which increases as Q_t falls. Trading down thus relaxes the rationing constraint precisely when it is most costly.

This inversion also complicates Jaimovich et al. (2019)'s empirical shortcut of using higher prices as a proxy for higher quality (see their Appendix A.3 on the NK model). In their setting, flexible prices transmit quality into the price schedule unambiguously. With nominal rigidities the mapping breaks, because a firm stuck at an inherited price may produce either high or low quality, and the household's quality choice responds to the *aggregate* rationing margin, not to price signals *per se*. The derivative I compute shows that the quality margin depends on the entire rationed supply distribution $\mathbb{E}_{f(r)|_0^f} [P(r)^{\frac{(1-\alpha)(\varepsilon-1)}{\alpha\varepsilon}}]$ and the semi-elasticity $\frac{Q'(Q)}{Q(Q)}$, neither of which maps cleanly onto a simple price-quality gradient. Empirically, this suggests that the trading-down channel may operate in the opposite direction than Jaimovich et al. (2019) document when rationing rather than income is the dominant constraint.

5 Conclusions

This paper develops a New Old Keynesian model in which firms optimally ration demand when selling at a price below marginal cost becomes unprofitable. Unlike standard New Keynesian models, where firms passively accommodate any level of demand, rationing introduces a capacity constraint that alters the transmission of shocks and the cyclical behavior of markups.

The model yields four main results. First, a positive demand shock forces low-price firms against their capacity wall, redirecting expenditure toward high-markup unrationed firms. Aggregate markups rise with measured output, matching the empirical findings of Nekarda and Ramey (2020), even though each firm's desired markup falls. This is a compositional effect: the expenditure share shifts toward high-margin varieties, not a change in individual pricing behavior. Second, the welfare-relevant shadow price decouples from other existing indices, typically exceeding them during rationing episodes. For instance, an expenditure-weighted index understates the true cost of living, generating a measurement illusion in which real consumption appears to rise while

welfare-relevant consumption falls. Third, price dispersion carries first-order welfare costs under rationing, in contrast to the standard model where these costs are second order. A mean-preserving spread of relative prices pushes additional goods into the rationed set without a proportional increase in their supply, directly reducing aggregate consumption. Fourth, an economy with rationing and price rigidities is capacity constrained and admits a finite maximum output. For sufficiently large demand shocks, no Walrasian equilibrium exists: desired consumption exceeds the physical capacity of the economy, and the traditional Euler equation holds as an inequality.

Two extensions reinforce the core mechanism. The labor wedge acquires a structural micro-foundation through the divergence between $\tilde{P}_t$ and $\check{P}_t$, offering a supply-side interpretation of the wedge that Chari et al. (2009) argued New Keynesian models treated as a residual. Last, non-homothetic quality choice inverts the standard “trading down” channel: during demand-driven booms, households accept lower quality to relax the capacity constraint of rationed firms, not because they are poorer.

Taken together, these results suggest that rationing is not a pathological corner case but a mechanism that reconciles the New Keynesian model with central empirical puzzles – procyclical markups, first-order costs of dispersion, and the possibility of non-Walrasian outcomes – without resorting to exogenous markup shocks or other *ad hoc* propositions. Important extensions of the methodology initiated in this paper are deferred to future research. These include the linearization of the model around trend-inflation steady states, the characterization of stochastic steady states, and an empirical validation of the mismeasurement hypothesis generated by this New Old Keynesian framework.

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Appendix: Mathematical Derivations

Derivation of Equation 4

Total expenditure is:

$$E_t = \int_{r \in R_t} P_t(r) \mathcal{Y}_t(r) + P_t^U \mathcal{U}_t^{\frac{\varepsilon}{\varepsilon-1}}$$

To find the effective shadow price of the aggregate consumption bundle (let's call it $\tilde{P}_t$), we need to determine the cost of purchasing one *marginal* unit of C_t . Since $\mathcal{R}_t$ is capped, all marginal adjustments must occur through $\mathcal{U}_t$. Differentiating the CES relationship $\mathcal{U}_t = C_t^{\frac{\varepsilon-1}{\varepsilon}} - \mathcal{R}_t$ from the previous equation yields:

$$\frac{d\mathcal{U}_t}{dC_t} = \frac{\varepsilon-1}{\varepsilon} C_t^{-\frac{1}{\varepsilon}}$$

The marginal cost of aggregate utility, $\tilde{P}_t$, is the derivative of total expenditure with respect to C_t . Since expenditure on rationed goods is locally fixed, we differentiate the unrationed expenditure component $E_t^U = P_t^U \mathcal{U}_t^{\frac{\varepsilon}{\varepsilon-1}}$ with respect to C_t using the chain rule:

$$\begin{aligned} \tilde{P}_t &= \frac{dE_t}{dC_t} \\ &= \frac{d}{d\mathcal{U}_t} \left(P_t^U \mathcal{U}_t^{\frac{\varepsilon}{\varepsilon-1}} \right) \frac{d\mathcal{U}_t}{dC_t} \\ &= \left(P_t^U \frac{\varepsilon}{\varepsilon-1} \mathcal{U}_t^{\frac{1}{\varepsilon-1}} \right) \left(\frac{\varepsilon-1}{\varepsilon} C_t^{-\frac{1}{\varepsilon}} \right) \end{aligned}$$

Simplifying this expression leaves us with the effective aggregate price level:

$$\begin{aligned} \tilde{P}_t &= P_t^U \mathcal{U}_t^{\frac{1}{\varepsilon-1}} C_t^{-\frac{1}{\varepsilon}} \\ \tilde{P}_t &= P_t^U \left(\frac{\mathcal{U}_t}{C_t^{\frac{\varepsilon-1}{\varepsilon}}} \right)^{\frac{1}{\varepsilon-1}} \\ \tilde{P}_t &= P_t^U \left(1 - \frac{\mathcal{R}_t}{C_t^{\frac{\varepsilon-1}{\varepsilon}}} \right)^{\frac{1}{\varepsilon-1}} \end{aligned}$$

Derivation of Equation 6

The derivation of time t choice variables can be shown through a Lagrangian optimization:

$$\mathcal{L} \equiv \sum_{j=0}^{\infty} \beta^j \mathbb{E}_t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} + \frac{\lambda_t}{\tilde{P}_t} \left(W_t N_t + B_t - \frac{B_{t+1}}{I_t} + \varpi_t - \tilde{P}_t C_t \right) + \xi_t (\mathcal{Y}_t^{\mathcal{F}} - C_t) \right]$$

with FOCs:

$$\begin{aligned}
[C_t] : \quad & \lambda_t = C_t^{-\sigma} - \xi_t \\
[N_t] : \quad & N_t^\varphi = \lambda_t \frac{W_t}{\tilde{P}_t} \\
[B_{t+1}] : \quad & \frac{\lambda_t}{\lambda_{t+1}} = \frac{I_t \beta}{\mathbb{E}_t \tilde{\Pi}_{t+1}}
\end{aligned}$$

which can be easily rearranged to obtain the optimality conditions.

Proof of Proposition 3.1

I split the proof into two cases. The former proves global continuity of $\tilde{P}_t$ for a continuously-differential price kernel f . The latter proves global continuity for the discrete “Calvo-ladder” distribution.

Let me start with a continuous price distribution f . Start with the structural definition of the shadow price, taking the natural log (denoted with lowercase letters), and computing its derivative:

$$\begin{aligned}
\tilde{P}_t &= P_t^U \left(\frac{\mathcal{U}_t}{C_t^{\frac{\varepsilon-1}{\varepsilon}}} \right)^{\frac{1}{\varepsilon-1}} \\
\tilde{p}_t &= p_t^U + \frac{1}{\varepsilon-1} u_t - \frac{1}{\varepsilon} c_t \\
\frac{\partial \tilde{p}_t}{\partial \underline{P}_t} &= \frac{\partial p_t^U}{\partial \underline{P}_t} + \frac{1}{\varepsilon-1} \frac{\partial u_t}{\partial \underline{P}_t} - \frac{1}{\varepsilon} \frac{\partial c_t}{\partial \underline{P}_t} \\
\frac{\partial \tilde{p}_t}{\partial \underline{P}_t} &= \frac{\partial p_t^U}{\partial \underline{P}_t} + \frac{1}{\varepsilon-1} \left(\frac{1}{\mathcal{U}_t} \right) \frac{\partial \mathcal{U}_t}{\partial \underline{P}_t} - \frac{1}{\varepsilon} \left(\frac{1}{C_t} \right) \frac{\partial C_t}{\partial \underline{P}_t} \\
\frac{\partial \tilde{p}_t}{\partial \underline{P}_t} &= \frac{\partial p_t^U}{\partial \underline{P}_t} + \frac{1}{\varepsilon-1} \left(\frac{1}{\mathcal{U}_t} \right) \frac{\partial \mathcal{U}_t}{\partial \underline{P}_t}
\end{aligned}$$

where $\frac{\partial C_t}{\partial \underline{P}_t} = 0$ since the price threshold only changes the partition while $\mathcal{U}_t + \mathcal{R}_t \equiv C_t^{\frac{\varepsilon-1}{\varepsilon}}$ for all t .

To evaluate $\frac{\partial \mathcal{U}_t}{\partial \underline{P}_t}$, let $f(P)$ denote the probability mass function for each price P . Applying the Leibniz rule to compute the derivative from the definition of $\mathcal{U}_t$ yields:

$$\begin{aligned}
\mathcal{U}_t &= C_t^{\frac{\varepsilon-1}{\varepsilon}} - \int_0^{\underline{P}_t} \mathcal{Y}_t(P)^{\frac{\varepsilon-1}{\varepsilon}} f(P) dP \\
\frac{\partial \mathcal{U}_t}{\partial \underline{P}_t} &= -Y_t(\underline{P}_t)^{\frac{\varepsilon-1}{\varepsilon}} f(\underline{P}_t) + \int_0^{\underline{P}_t} \frac{\partial}{\partial \underline{P}_t} Y_t(P)^{\frac{\varepsilon-1}{\varepsilon}} f(P) dP \\
\frac{\partial \mathcal{U}_t}{\partial \underline{P}_t} &= -Y_t(\underline{P}_t)^{\frac{\varepsilon-1}{\varepsilon}} f(\underline{P}_t)
\end{aligned}$$

Next, I evaluate the term $\frac{\partial p_t^U}{\partial \underline{P}_t}$. Isolating the unrationed price index integral by defining $I_t \equiv \int_{\underline{P}_t}^{P_t^*} P^{1-\varepsilon} f(P) dP$,

the unrationed price sub-index becomes $P_t^U = I_t^{\frac{1}{1-\varepsilon}}$. Applying the chain rule to $p_t^U = \frac{1}{1-\varepsilon} \ln I_t$ gives:

$$\begin{aligned}\frac{\partial p_t^U}{\partial P_t} &= \frac{1}{1-\varepsilon} \frac{1}{I_t} \frac{\partial I_t}{\partial P_t} \\ \frac{\partial I_t}{\partial P_t} &= -(P_t)^{1-\varepsilon} f(P_t) \\ \frac{\partial p_t^U}{\partial P_t} &= \frac{1}{\varepsilon-1} \frac{(P_t)^{1-\varepsilon} f(P_t)}{I_t}\end{aligned}$$

Plugging these terms back into the derivative of the welfare-relevant price index:

$$\begin{aligned}\frac{\partial \tilde{p}_t}{\partial P_t} &= \frac{\partial p_t^U}{\partial P_t} + \frac{1}{\varepsilon-1} \left(\frac{1}{\mathcal{U}_t} \right) \frac{\partial \mathcal{U}_t}{\partial P_t} \\ &= \frac{1}{\varepsilon-1} \left[\frac{(P_t)^{1-\varepsilon} f(P_t)}{I_t} - \frac{\mathcal{Y}_t(P_t)^{\frac{\varepsilon-1}{\varepsilon}} f(P_t)}{\mathcal{U}_t} \right] \\ &= \frac{f(P_t)}{\varepsilon-1} \left[\frac{(P_t)^{1-\varepsilon}}{I_t} - \frac{(P_t)^{1-\varepsilon}}{I_t} \right] = 0\end{aligned}$$

where the final substitution follows from the definition of maximum profitable supply, $\mathcal{Y}_t(P_t) = \left(\frac{P_t}{P_t^U} \right)^{-\varepsilon} \mathcal{U}_t^{\frac{\varepsilon-1}{\varepsilon}}$, which implies $\frac{\mathcal{Y}_t(P_t)^{\frac{\varepsilon-1}{\varepsilon}}}{\mathcal{U}_t} = \left(\frac{P_t}{P_t^U} \right)^{1-\varepsilon} = \frac{P_t^{1-\varepsilon}}{I_t}$.

I will continue the proof by specifying a useful case of non-globally-differentiable pricing kernel: the discrete Calvo ladder. Expanding the utility structure based on a discrete Calvo-ladder partition gives:

$$\begin{aligned}\mathcal{U}_t &= C_t^{\frac{\varepsilon-1}{\varepsilon}} - \mathcal{R}_t \\ &= \left[(1-\theta) \sum_{k=0}^{j^*-1} \theta^k \mathcal{D}_t(P_t(k))^{\frac{\varepsilon-1}{\varepsilon}} + (1-\theta)\theta^{j^*} \sum_{k=0}^{\infty} \theta^k \mathcal{Y}_t(P_t(j^*+k))^{\frac{\varepsilon-1}{\varepsilon}} \right] - (1-\theta)\theta^{j^*} \sum_{k=0}^{\infty} \theta^k \mathcal{Y}_t(P_t(j^*+k))^{\frac{\varepsilon-1}{\varepsilon}}\end{aligned}$$

Suppose the economy sits at a boundary where unconstrained demand equals maximum profitable supply for the threshold cohort, that is $\mathcal{D}_t(P_t(j^*-1)) = \mathcal{Y}_t(P_t(j^*-1))$. If the rationing threshold expands such that the last unrationed cohort shifts from j^*-1 to j^*-2 , the total consumption index $C_t^{\frac{\varepsilon-1}{\varepsilon}}$ remains identical because the boundary mass carries the exact same quantity across partitions. The discrete change in $\mathcal{U}_t$ is:

$$\begin{aligned}\mathcal{U}_t|_{j=j^*-2}^{j=j^*-1} &= 0 - \left((1-\theta)\theta^j \sum_{k=0}^{\infty} \theta^k \mathcal{Y}_t(P_t(j+k))^{\frac{\varepsilon-1}{\varepsilon}} \right) \Big|_{j=j^*-2}^{j=j^*-1} \\ &= 0 - (1-\theta) \left[\theta^{j^*-1} \sum_{k=0}^{\infty} \theta^k \mathcal{Y}_t(P_t(j^*-1+k))^{\frac{\varepsilon-1}{\varepsilon}} - \theta^{j^*-2} \sum_{k=0}^{\infty} \theta^k \mathcal{Y}_t(P_t(j^*-2+k))^{\frac{\varepsilon-1}{\varepsilon}} \right] \\ &= -(1-\theta)\theta^{j^*-2} \mathcal{Y}_t(P_t(j^*-2))^{\frac{\varepsilon-1}{\varepsilon}}\end{aligned}$$

which matches the discrete counterpart of the continuous derivative $-\mathcal{Y}_t(P_t)^{\frac{\varepsilon-1}{\varepsilon}} f(P_t)$. For the discrete

unrationed price sub-index component, we define I_t :

$$I_t = (1 - \theta) \sum_{k=0}^{j^*-1} \theta^k P_t(k)^{1-\varepsilon}$$

Shifting the threshold to $j^* - 2$ moves cohort $j^* - 1$ into the rationed set, yielding the discrete displacement:

$$\begin{aligned} I_t|_{j=j^*-2}^{j=j^*-1} &= (1 - \theta) \sum_{k=0}^{j^*-2} \theta^k P_t(k)^{1-\varepsilon} - (1 - \theta) \sum_{k=0}^{j^*-1} \theta^k P_t(k)^{1-\varepsilon} \\ &= -(1 - \theta) \theta^{j^*-1} P_t(j^* - 1)^{1-\varepsilon} \end{aligned}$$

Evaluating the corresponding ratio for the unrationed price index yields:

$$\begin{aligned} \frac{P_t^U(j^* - 2)}{P_t^U(j^* - 1)} &= \left(\frac{I_t(j^* - 2)}{I_t(j^* - 1)} \right)^{\frac{1}{1-\varepsilon}} \\ &= \left(\frac{I_t(j^* - 1) - (1 - \theta) \theta^{j^*-1} P_t(j^* - 1)^{1-\varepsilon}}{I_t(j^* - 1)} \right)^{\frac{1}{1-\varepsilon}} \\ &= \left(1 - \frac{(1 - \theta) \theta^{j^*-1} P_t(j^* - 1)^{1-\varepsilon}}{I_t(j^* - 1)} \right)^{\frac{1}{1-\varepsilon}} \end{aligned}$$

To prove the level of the shadow price index $\tilde{P}_t$ remains continuous across this discrete boundary, we utilize the relationship $\tilde{P}_t^{1-\varepsilon} = C_t^{\frac{\varepsilon-1}{\varepsilon}} \frac{I_t}{\mathcal{U}_t}$. Since the marginal cohort $j^* - 1$ is exactly constrained at the boundary:

$$\begin{aligned} \mathcal{Y}_t(P_t(j^* - 1)) &= \left(\frac{P_t(j^* - 1)}{P_t^U(j^* - 1)} \right)^{-\varepsilon} \mathcal{U}_t(j^* - 1)^{\frac{\varepsilon}{\varepsilon-1}} \\ \mathcal{Y}_t(P_t(j^* - 1))^{\frac{\varepsilon-1}{\varepsilon}} &= \frac{P_t(j^* - 1)^{1-\varepsilon}}{I_t(j^* - 1)} \mathcal{U}_t(j^* - 1) \\ \frac{P_t(j^* - 1)^{1-\varepsilon}}{\mathcal{Y}_t(P_t(j^* - 1))^{\frac{\varepsilon-1}{\varepsilon}}} &= \frac{I_t(j^* - 1)}{\mathcal{U}_t(j^* - 1)} \\ \frac{(1 - \theta) \theta^{j^*-1} P_t(j^* - 1)^{1-\varepsilon}}{(1 - \theta) \theta^{j^*-1} \mathcal{Y}_t(P_t(j^* - 1))^{\frac{\varepsilon-1}{\varepsilon}}} &= \frac{I_t(j^* - 1)}{\mathcal{U}_t(j^* - 1)} \end{aligned}$$

Using the algebraic property that if $\frac{a}{b} = \frac{c}{d}$, then $\frac{a-c}{b-d} = \frac{a}{b}$, we subtract the cohort's components from the indices of the $j^* - 1$ regime to establish the ratio under the $j^* - 2$ regime:

$$\frac{I_t(j^* - 2)}{\mathcal{U}_t(j^* - 2)} = \frac{I_t(j^* - 1) - (1 - \theta) \theta^{j^*-1} P_t(j^* - 1)^{1-\varepsilon}}{\mathcal{U}_t(j^* - 1) - (1 - \theta) \theta^{j^*-1} \mathcal{Y}_t(P_t(j^* - 1))^{\frac{\varepsilon-1}{\varepsilon}}} = \frac{I_t(j^* - 1)}{\mathcal{U}_t(j^* - 1)}$$

This directly implies:

$$\tilde{P}_t(j^* - 2)^{1-\varepsilon} = C_t^{\frac{\varepsilon-1}{\varepsilon}} \frac{I_t(j^* - 2)}{\mathcal{U}_t(j^* - 2)} = C_t^{\frac{\varepsilon-1}{\varepsilon}} \frac{I_t(j^* - 1)}{\mathcal{U}_t(j^* - 1)} = \tilde{P}_t(j^* - 1)^{1-\varepsilon}$$

Thus, $\tilde{P}_t(j^* - 2) = \tilde{P}_t(j^* - 1)$, completing the proof of global continuity for both the continuous distribution and the discrete Calvo-ladder.

Proof of Proposition 3.2

Recall the previous foundations, and define relative prices and price dispersion as $\wp_t(i) \equiv \frac{P_t(i)}{\bar{P}_t}$ and $\Delta_t \equiv \int_0^1 \wp_t(i) di$ respectively.

In the standard model without rationing, all goods belong to U_t . Around a symmetric equilibrium where all $\wp_t(i) = 1$, a second-order Taylor expansion yields:

$$\mathcal{Y}_t - Y_t \approx \frac{\varepsilon}{2} Y_t \text{Var}[\ln \wp_t(i)] + \mathcal{O}(\|\ln \wp\|^3)$$

Hence the wedge is second order, and a marginal increase in price dispersion leaves output unchanged to a first approximation.

Now restore the rationed set R_t . Suppose the economy is in a situation where a positive measure of goods are rationed. We can write the linear sum as:

$$\mathcal{Y}_t = Y_t \int_{u \in U_t} \wp_t(u)^{-\varepsilon} + \int_{r \in R_t} \mathcal{Y}_t(r)$$

Using the threshold condition to replace $\mathcal{Y}_t(r)$ for the rationed goods that are exactly at the boundary, and noting that for all rationed goods $\mathcal{Y}_t(r) < \wp_t(r)^{-\varepsilon} Y_t$ — that is, they would be demanded more if unrationed — we can subtract Y_t from both sides, and by adding and subtracting the notional demand of the rationed goods, $Y_t \int_{r \in R_t} \wp_t(r)^{-\varepsilon}$, obtain:

$$\begin{aligned} \mathcal{Y}_t - Y_t &= Y_t \int_{u \in U_t} \wp_t(u)^{-\varepsilon} + \int_{r \in R_t} \mathcal{Y}_t(r) - Y_t \\ &= \int_{r \in R_t} \mathcal{Y}_t(r) - Y_t \int_{r \in R_t} \wp_t(r)^{-\varepsilon} + Y_t \int_{u \in U_t} \wp_t(u)^{-\varepsilon} + Y_t \int_{r \in R_t} \wp_t(u)^{-\varepsilon} - Y_t \\ &= \underbrace{\int_{r \in R_t} \mathcal{Y}_t(r) - \wp_t(r)^{-\varepsilon} Y_t}_{<0} + Y_t \left(\int_{u \in U_t} \wp_t(u)^{-\varepsilon} - 1 \right) \end{aligned}$$

The second bracket is the familiar standard NK term, which is second order around symmetry. The first bracket, however, is strictly negative and proportional to the mass of rationed goods $|R_t|$: It is first order in the extent of rationing. Therefore, any change that increases price dispersion pushing more goods into the rationed set or lowering their relative price further widens the negative wedge at a first-order rate.

Derivation of Equation 8

To derive the true optimal pricing rule, one must set up the piecewise optimization problem where the firm's profit inside the rationed and unrationed states is a function of its price, $\varpi^R(P_t^*)$. I will apply the Leibniz rule to handle the moving boundaries across rationing regions, evaluate the interior marginal profits of both regimes, and solve for the reset price P_t^* .

Let $\mathcal{X}^U(P_t^*)$ be the set of future macroeconomic states where the firm remains unrationed, and $\mathcal{X}^R(P_t^*)$ be the set of states where the capacity constraint binds. The firm's expected value function is:

$$V_t(P_t^*) = \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} \left[\int_{\mathcal{X}^U} \varpi^U(P_t^*, X) f(X) dX + \int_{\mathcal{X}^R} \varpi^R(P_t^*, X) f(X) dX \right]$$

where $\Lambda_{t,t+k}$ represents the stochastic discount factor. Taking the first derivative with respect to P_t^* using the Leibniz integral rule yields the interior derivatives plus the boundary term evaluated at the threshold $\partial\mathcal{X}$:

$$\frac{\partial V_t}{\partial P_t^*} = \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} \left\{ \int_{\mathcal{X}^U} \frac{\partial \varpi^U}{\partial P_t^*} f(X) dX + \int_{\mathcal{X}^R} \frac{\partial \varpi^R}{\partial P_t^*} f(X) dX + \int_{\partial\mathcal{X}} [\varpi^U - \varpi^R] f(X) \frac{\partial \mathcal{X}}{\partial P_t^*} dX \right\}$$

By definition, the boundary $\partial\mathcal{X}$ is the exact state where notional demand perfectly equals the firm's maximum profitable capacity. Because physical output and price are identical across the boundary, $\Pi^U = \Pi^R$. The bracketed boundary term thus cancels. The firm's optimization restricts to the interior marginal profits:

$$\sum_{k=0}^{\infty} (\beta\theta)^k \mathbb{E}_t \left[\Lambda_{t,t+k} \left(\mathbb{1}_{t+k}^U \frac{\partial \varpi_{t+k}^U}{\partial P_t^*} + \mathbb{1}_{t+k}^R \frac{\partial \varpi_{t+k}^R}{\partial P_t^*} \right) \right] = 0$$

Where $\mathbb{1}^U$ and $\mathbb{1}^R$ are the indicator functions for the unrationed and rationed states. As can be seen, the marginal impact of the resetting price on the indicators is not part of the equation. The firm *does* change the probability of being rationed when it changes P_t^* ; however, the *marginal* impact on expected profit from crossing that boundary is exactly zero, and the firm is perfectly indifferent between being unrationed or rationed on the exact margin. The effect of the moving boundary drops out of the FOC entirely.

Inside the rationed state, the firm produces at its capacity limit $\mathcal{Y}_{t+k|t}$. Profit is $\varpi^R = P_t^* \mathcal{Y} - TC(\mathcal{Y})$. Taking the partial derivative with respect to the reset price:

$$\begin{aligned} \frac{\partial \varpi^R}{\partial P_t^*} &= \mathcal{Y} + P_t^* \frac{\partial \mathcal{Y}}{\partial P_t^*} - MC(\mathcal{Y}) \frac{\partial \mathcal{Y}}{\partial P_t^*} \\ \frac{\partial \varpi^R}{\partial P_t^*} &= \mathcal{Y} + P_t^* \frac{\partial \mathcal{Y}}{\partial P_t^*} \left(1 - \frac{1}{\underline{M}} \right) \\ \frac{\partial \varpi^R}{\partial P_t^*} &= \mathcal{Y}_{t+k|t} \left[1 + \frac{1-\alpha}{\alpha} \left(\frac{\underline{M}-1}{\underline{M}} \right) \right] \end{aligned}$$

where I took the derivative of the maximum profitable supply with respect to prices, obtained $\frac{\partial \mathcal{Y}}{\partial P_t^*} = \frac{1-\alpha}{\alpha} \frac{\mathcal{Y}}{P_t^*}$ and substituted this back into the marginal profit equation. Because $\alpha \in (0, 1)$ and $\underline{M} \geq 1$, the term in the brackets is a strictly positive constant. Let us define this constant as the *marginal capacity value of price*, $\mathcal{W} \geq 1^{**}$.

$$\frac{\partial \varpi^R}{\partial P_t^*} = \mathcal{W} \mathcal{Y}_{t+k|t}$$

In the unrationed state, the firm faces standard Dixit-Stiglitz demand $Y_{t+k|t}$, with elasticity $-\varepsilon$.

$$\begin{aligned} \varpi^U &= P_t^* Y - TC(Y) \\ \frac{\partial \varpi^U}{\partial P_t^*} &= Y + P_t^* \left(-\varepsilon \frac{Y}{P_t^*} \right) - MC^U \left(-\varepsilon \frac{Y}{P_t^*} \right) \\ \frac{\partial \varpi^U}{\partial P_t^*} &= Y_{t+k|t} \left[1 - \varepsilon + \varepsilon \frac{MC_{t+k|t}^U}{P_t^*} \right] \end{aligned}$$

I now plug both marginal profit expressions back into the stochastic first-order condition:

$$\begin{aligned}
& \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} \left[\mathbb{1}_{t+k}^U Y_{t+k|t} \left(1 - \varepsilon + \varepsilon \frac{MC_{t+k|t}^U}{P_t^*} \right) + \mathbb{1}_{t+k}^R \mathcal{WY}_{t+k|t} \right] = 0 \\
& \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} \left[\mathbb{1}_{t+k}^U Y_{t+k|t} \left(P_t^* (1 - \varepsilon) + \varepsilon MC_{t+k|t}^U \right) + \mathbb{1}_{t+k}^R P_t^* \mathcal{WY}_{t+k|t} \right] = 0 \\
& \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} \mathbb{1}^U Y (1 - \varepsilon) P_t^* + \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} \mathbb{1}^U Y \varepsilon MC^U + \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} \mathbb{1}^R \mathcal{WY} P_t^* = 0 \\
& P_t^* \left[(\varepsilon - 1) \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} \mathbb{1}_{t+k}^U Y_{t+k|t} - \mathcal{W} \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} \mathbb{1}_{t+k}^R \mathcal{Y}_{t+k|t} \right] = \varepsilon \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Lambda_{t,t+k} \mathbb{1}_{t+k}^U Y_{t+k|t}
\end{aligned}$$

The pricing rule follows.

Proof of Theorem 3.3

Proof. The claim is proved for a two-period economy, and then extended to the infinite horizon economy. The core mechanism will be the capacity collapse via the real wage.

At time 0, a mass $\theta \in (0, 1)$ of firms have a preset nominal price P_{old} (inherited from the steady state). The complementary mass $1 - \theta$ can choose any price for the two periods of activity (the price remains in effect for $t = 0, 1$). At time 1, the economy returns to the steady state (no shock, all prices flexible). Hence the period-1 aggregates are known constants, and all firms will be unrationed. The only interesting decision is the price chosen by the free firms at time 0.

An equilibrium consists of

$$\{C_t, N_t, W_t, \tilde{P}_t\}_{t=0,1} \quad \{P_{ss}, P^*\}$$

and the partition of goods into a rationed set R and an unrationed set U , such that Euler and Labor supply equations hold, goods market clears, $C_t = Y_t$ where Y_t is the split-CES aggregate of the $Y_t(i)$; labor market clears $N_t = \int_0^1 (Y_t(i)/A)^{1/\alpha} di$; rationing partitions hold, that is $Y_t(r) = \mathcal{Y}_t(r) < (P_t(r)/\tilde{P}_t)^{-\varepsilon} Y_t$ and $Y_t(u) = (P_t(u)/\tilde{P}_t)^{-\varepsilon} Y_t \leq \mathcal{Y}_t(u)$; and flexible-price firms choose P^* to maximize their total discounted profit over periods 0 and 1 taking aggregate variables as given.

Let the real wage be $V_t \equiv W_t/\tilde{P}_t$, where $\tilde{P}_t$ is the welfare-relevant shadow price index. At $t = 0$ we suppress the time index for all variables. Total capacity (linear sum over all firms) is

$$\mathcal{Y}(V) = \int_0^1 \mathcal{Y}(i) di = \underbrace{A^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{\mathcal{M}} \right)^{\frac{\alpha}{1-\alpha}}}_{\equiv \Phi} V^{-\frac{\alpha}{1-\alpha}} \int_0^1 \wp(i)^{\frac{\alpha}{1-\alpha}} di$$

Because relative prices are bounded, the integral $\int_0^1 \wp(i)^{\frac{\alpha}{1-\alpha}} di$ is finite and independent of V . Thus:

$$\mathcal{Y}(V) \propto V^{-\frac{\alpha}{1-\alpha}} \implies \mathcal{Y}(V) \xrightarrow{V \rightarrow \infty} 0$$

Equilibrium output Y cannot exceed total capacity; labour input is $N = \int_0^1 (Y(i)/A)^{1/\alpha} di$. Since

$Y(i) \leq \mathcal{Y}(i)$ for every i ,

$$\begin{aligned} N &\leq \int_0^1 (\mathcal{Y}(i)/A)^{1/\alpha} di \\ &\leq \Phi' V^{-\frac{1}{1-\alpha}} \int_0^1 \wp(i)^{\frac{1}{1-\alpha}} di \equiv \bar{N}(V). \end{aligned}$$

Again, $\bar{N}(V) \xrightarrow{V \rightarrow \infty} 0$. Solve for Y :

$$Y \leq V^{\frac{1}{\sigma}} N^{-\frac{\varphi}{\sigma}} \leq V^{\frac{1}{\sigma}} \bar{N}(V)^{-\frac{\varphi}{\sigma}}$$

Define the feasible output correspondence:

$$Y^{\mathcal{F}}(V) \equiv \min \left\{ \mathcal{Y}(V), V^{\frac{1}{\sigma}} \bar{N}(V)^{-\frac{\varphi}{\sigma}} \right\}$$

Because $\mathcal{Y}(V) \propto V^{-\frac{\alpha}{1-\alpha}}$ and $\bar{N}(V) \propto V^{-\frac{1}{1-\alpha}}$, the following remarks hold:

$$\begin{aligned} V \rightarrow 0 &\implies \begin{cases} \mathcal{Y}(V) \rightarrow \infty \\ V^{\frac{1}{\sigma}} \bar{N}(V)^{-\frac{\varphi}{\sigma}} \rightarrow 0 \end{cases} \quad \text{since } N(V)^{-\frac{\varphi}{\sigma}} \bar{N}(V)^{-\frac{\varphi}{\sigma}} \propto V^{\frac{1}{\sigma} + \frac{\varphi}{\sigma(1-\alpha)}} \\ V \rightarrow \infty &\implies \mathcal{Y}(V) \rightarrow 0 \end{aligned}$$

As V goes to infinity, the total capacity shrinks and dominates the minimum. As V goes to zero, the labor supply equation shrinks and dominates the minimum. Hence, $Y^{\mathcal{F}}(V)$ is continuous, positive on $(0, \infty)$, and tends to 0 at both extremes. Therefore, by the Stone-Weierstrass theorem it attains a finite maximum:

$$Y_{\max} \equiv \max_{V>0} Y^{\mathcal{F}}(V) < \infty$$

and any equilibrium must satisfy $Y \leq Y_{\max}$.

The Euler equation links Y to the future steady-state output Y_{ss} :

$$\begin{aligned} Y &= (\beta R)^{-\frac{1}{\sigma}} Y_{ss} \\ R &= \frac{1+v}{\beta} \end{aligned}$$

A negative demand shock $v < 0$ lowers the denominator, so $Y > Y_{ss}$. By choosing $|v|$ sufficiently large, we obtain $Y > Y_{\max}$. The capacity bound and the labour-supply bound together imply $Y_0 \leq Y^{\mathcal{F}}(V) \leq Y_{\max}$ for any equilibrium real wage. Yet, the Euler equation can demand $Y > Y_{\max}$. For such a demand shock, the Walrasian equilibrium—defined by the Euler equation holding as an equality—would require output to exceed the economy's physical capacity, which is impossible. Therefore a Walrasian equilibrium does not exist.

Beyond this economy-wide capacity constraint, the economy enters a non-Walrasian regime, and the household's optimization problem must include the (binding) quantity constraint $C_t \leq \mathcal{Y}_t$. Introducing a non-negative Lagrange multiplier ξ_t on this constraint modifies the intertemporal optimality condition to

$$C_t^{-\sigma} = \beta R_{t+1} C_{t+1}^{-\sigma} + \xi_t \tilde{P}_t, \quad \xi_t \geq 0.$$

and with $C_t = Y_t$ and $C_{t+1} = Y_{ss}$, this yields

$$Y^{-\sigma} = \beta R Y_{ss}^{-\sigma} + \xi \tilde{P}, \quad \xi > 0 \iff Y = Y_{\max}.$$

The standard Euler equation is thus replaced by an inequality: $Y^{-\sigma} > \beta R Y_{ss}^{-\sigma}$; the multiplier ξ_t measures the shadow value of the binding capacity constraint, that is the wedge between the marginal utility of consumption today and the discounted marginal utility of tomorrow. In this constrained equilibrium, markets still clear (output equals consumption, labour equals demand), households optimize given the constraint, and firms still satisfy their optimality conditions. The equilibrium is non-Walrasian precisely because the intertemporal first-order condition contains a positive shadow price of the aggregate shortage.

The finite-horizon argument extends naturally to an infinite-horizon setting with staggered pricing and a possibly time-varying real rate path. The reasoning below retains the same structural ingredients as the two-period case.

Because the distribution of relative prices $\wp_t(i)$ is determined by a Calvo mechanism (or any bounded support), the integrals in $\mathcal{Y}_t(i)$ and the corresponding one for $\bar{N}_t$ are bounded away from zero and infinity. Hence, for every t , there exists a finite, time-varying maximum feasible output

$$Y_t^{\mathcal{F}} \equiv \max_{V_t > 0} \min \left\{ \mathcal{Y}_t(V_t), V_t^{\frac{1}{\sigma}} (\bar{N}_t(V_t))^{-\frac{\varphi}{\sigma}} \right\} < \infty$$

Any Walrasian equilibrium must therefore satisfy $Y_t \leq Y_t^{\mathcal{F}}$ for all t .

Now consider the intertemporal Euler equation (no uncertainty, for simplicity):

$$C_t^{-\sigma} = \beta R_{t+1} C_{t+1}^{-\sigma},$$

where R_t is the gross real interest rate between $t-1$ and t . Goods-market clearing $Y_t = C_t$ yields

$$Y_t = (\beta R_{t+1})^{-\frac{1}{\sigma}} Y_{t+1}$$

$$Y_0 = Y_T \prod_{k=1}^T (\beta R_k)^{-\frac{1}{\sigma}}$$

If the economy eventually returns to a steady state with output $Y_{ss} < \infty$, then $Y_T \rightarrow Y_{ss}$ as $T \rightarrow \infty$, so

$$Y_0 = Y_{ss} \lim_{T \rightarrow \infty} \prod_{k=1}^T (\beta R_k)^{-\frac{1}{\sigma}}$$

A (persistent) demand expansion (real rates below $1/\beta$) makes $\beta R_k < 1$ for $k \geq 1$, causing the product to diverge as $T \rightarrow \infty$. For instance, if $R_k = \frac{1+v_k}{\beta}$ with $v_k = \rho_v^k v$ for $v < 0$, the product is $\prod_{k=1}^{\infty} (1 + \rho^k v)^{-\frac{1}{\sigma}} \xrightarrow{v \rightarrow -1^+} \infty$, which grows without bound. Therefore, for any finite capacity sequence $\{Y_t^{\mathcal{F}}\}$, one can choose v sufficiently close to -1 so that the unconstrained Euler equation demands $Y_0 > Y_0^{\mathcal{F}}$, $Y_1 > Y_1^{\mathcal{F}}$, *et cetera*.

Consequently, the Walrasian equilibrium with equality in the Euler equation cannot exist. Instead, the economy enters a non-Walrasian path in which the aggregate capacity constraint binds at every date where

the unconstrained demand would violate the bound. The households intertemporal condition becomes

$$Y_t^{-\sigma} = \beta R_{t+1} Y_{t+1}^{-\sigma} + \xi_t \tilde{P}_t, \quad \xi_t \geq 0,$$

with $\xi_t > 0$ whenever the bound is binding. The sequence of multipliers $\{\xi_t\}_{t=0}^{\infty}$ absorbs the excess notional demand, guaranteeing that realized output equals the feasible maximum and that the goods market clears. This is precisely the non-Walrasian equilibrium characterized by a positive shadow value of the capacity constraint. □

Proof of Proposition 3.4

At the boundary $P = P_0$, the firm is exactly at the threshold where $\mathcal{D} = \mathcal{Y}$. The left derivative (approaching from the rationed side) is $\mathcal{WY}_0 > 0$. The right derivative (approaching from the unrationed side) includes the future terms. If the FOC sum evaluated at P_0 (approaching from the rationed side) is positive, the firm wants to raise its price — escaping rationing. If it is negative, the firm optimally stays in the rationed region. The condition stated is precisely $\frac{\partial V}{\partial P}|_{P \uparrow P_0} < 0$, which implies the optimal P^* lies strictly below P_0 .

Derivation of Equation 18

The derivation is a pure algebraic effort:

$$C^{\frac{\varepsilon-1}{\varepsilon}} = \int_{u \in U} C(u)^{\frac{\varepsilon-1}{\varepsilon}} du + \int_0^f \left[\left(\frac{A}{Q(Q)} \right)^{\frac{1}{\alpha}} \left(\frac{\alpha P(r)}{\underline{MW}} \right)^{\frac{1-\alpha}{\alpha}} \right]^{\frac{\varepsilon-1}{\varepsilon}} f(r) dr$$

$$\frac{\partial C^{\frac{\varepsilon-1}{\varepsilon}}}{\partial Q} = \frac{\partial}{\partial Q} \int_0^f \left[\left(\frac{A}{Q(Q)} \right)^{\frac{1}{\alpha}} \left(\frac{\alpha P(r)}{\underline{MW}} \right)^{\frac{1-\alpha}{\alpha}} \right]^{\frac{\varepsilon-1}{\varepsilon}} f(r) dr$$

The unrationed term is independent of Q , while Leibniz boundary terms vanish by the envelope. Then:

$$\begin{aligned}
&= \int_0^f \frac{\partial}{\partial Q} \left[\left(\frac{A}{Q(Q)} \right)^{\frac{1}{\alpha}} \left(\frac{\alpha P(r)}{\underline{MW}} \right)^{\frac{1-\alpha}{\alpha}} \right]^{\frac{\varepsilon-1}{\varepsilon}} f(r) dr \\
&= \int_0^f \frac{\partial}{\partial Q} \left[\left(\frac{A}{Q(Q)} \right)^{\frac{\varepsilon-1}{\alpha\varepsilon}} \left(\frac{\alpha P(r)}{\underline{MW}} \right)^{\frac{(1-\alpha)(\varepsilon-1)}{\alpha\varepsilon}} \right] f(r) dr \\
&= \int_0^f \left(\frac{\alpha P(r)}{\underline{MW}} \right)^{\frac{(1-\alpha)(\varepsilon-1)}{\alpha\varepsilon}} \cdot \frac{\partial}{\partial Q} \left(\frac{A}{Q(Q)} \right)^{\frac{\varepsilon-1}{\alpha\varepsilon}} f(r) dr \\
&= \int_0^f \left(\frac{\alpha P(r)}{\underline{MW}} \right)^{\frac{(1-\alpha)(\varepsilon-1)}{\alpha\varepsilon}} \cdot \left[\frac{\varepsilon-1}{\alpha\varepsilon} \left(\frac{A}{Q(Q)} \right)^{\frac{\varepsilon-1}{\alpha\varepsilon}-1} \cdot A \cdot \left(-\frac{Q'(Q)}{Q(Q)^2} \right) \right] f(r) dr \\
&= -\frac{\varepsilon-1}{\alpha\varepsilon} \left(\frac{A}{Q(Q)} \right)^{\frac{\varepsilon-1}{\alpha\varepsilon}} \frac{Q'(Q)}{Q(Q)} \int_0^f \left(\frac{\alpha P(r)}{\underline{MW}} \right)^{\frac{(1-\alpha)(\varepsilon-1)}{\alpha\varepsilon}} f(r) dr \\
&= -\frac{\varepsilon-1}{\alpha\varepsilon} \left(\frac{A}{Q(Q)} \right)^{\frac{\varepsilon-1}{\alpha\varepsilon}} \frac{Q'(Q)}{Q(Q)} \left(\frac{\alpha}{\underline{MW}} \right)^{\frac{(1-\alpha)(\varepsilon-1)}{\alpha\varepsilon}} \int_0^f P(r)^{\frac{(1-\alpha)(\varepsilon-1)}{\alpha\varepsilon}} f(r) dr \\
&= -\frac{\varepsilon-1}{\alpha\varepsilon} \left(\frac{A}{Q(Q)} \right)^{\frac{\varepsilon-1}{\alpha\varepsilon}} \frac{Q'(Q)}{Q(Q)} \left(\frac{\alpha}{\underline{MW}} \right)^{\frac{(1-\alpha)(\varepsilon-1)}{\alpha\varepsilon}} \mathbb{E}_{f(r)|_0^f} \left[P(r)^{\frac{(1-\alpha)(\varepsilon-1)}{\alpha\varepsilon}} \right]
\end{aligned}$$